

# Different observables protect different pasts

Changing the validation target changes the memory mixture and the best fixed transition rule.

| validation target                                                                            | velocity | archive | learned | pairs | best fixed      | interpretation                    |
|----------------------------------------------------------------------------------------------|----------|---------|---------|-------|-----------------|-----------------------------------|
|  balanced    | 0.44     | 0.12    | 0.44    | 0.00  | <b>velocity</b> | no single past is enough          |
|  arrival     | 0.38     | 0.12    | 0.44    | 0.06  | <b>velocity</b> | arrival tolerates velocity memory |
|  dilution    | 0.38     | 0.06    | 0.56    | 0.00  | <b>learned</b>  | context helps plume structure     |
|  pairs       | 0.25     | 0.12    | 0.62    | 0.00  | <b>archive</b>  | pairs ask for different history   |
|  encounters  | 0.25     | 0.12    | 0.62    | 0.00  | <b>velocity</b> | encounters still favor velocity   |
|  no reaction | 0.44     | 0.12    | 0.44    | 0.00  | <b>velocity</b> | velocity remains economical       |

 velocity

 archive

 learned

 pairs